

0.1. Recap

- **Expectation**/mean of a random variable R is denoted by $\mathbb{E}[R]$ and “summarizes” its distribution. Formally, $\mathbb{E}[R] := \sum_{\omega \in S} \Pr[\omega] R[\omega]$.
 - An alternate definition is $\mathbb{E}[R] = \sum_{x \in \text{Range}(R)} x \Pr[R = x]$.
- **Linearity of expectation**: for n random variables $R_1, R_2, \dots, R_n$ and constants $a_1, a_2, \dots, a_n$, $\mathbb{E}\left[\sum_{i=1}^n a_i R_i\right] = \sum_{i=1}^n a_i \mathbb{E}[R_i]$.
- **Expectation for common distributions**:
 - If $R \sim \text{Bernoulli}(p)$, $\mathbb{E}[R] = p$.
 - If $R \sim \text{Uniform}(\{v_1, \dots, v_n\})$, $\mathbb{E}[R] = \frac{v_1 + \dots + v_n}{n}$.
 - If $R \sim \text{Binomial}(n, p)$, $\mathbb{E}[R] = np$.
 - If $R \sim \text{Geometric}(p)$, $\mathbb{E}[R] = \frac{1}{p}$.

1. Conditional Expectation

Similarly to probabilities, expectations can be conditioned on some event.

- **Definition**: For random variable R , the expected value of R conditioned on an event A is:

$$\mathbb{E}[R|A] := \sum_{x \in \text{Range}(R)} x \Pr[R = x|A]$$

Q: If we throw a standard dice and define R to be the random variable equal to the number that comes up, what is the expected value of R given that the number is at most 4?

Let A be the event that the number is at most 4 — $A = \{1, 2, 3, 4\}$.

$$\Pr[R = 1|A] = \frac{\Pr[(R=1) \cap A]}{\Pr[A]} = \frac{\Pr[R=1]}{\Pr[A]} = \frac{\frac{1}{6}}{\frac{4}{6}} = \frac{1}{4}$$

Similarly, $\Pr[R = 2|A] = \Pr[R = 3|A] = \Pr[R = 4|A] = \frac{1}{4}$ and $\Pr[R = 5|A] = \Pr[R = 6|A] = 0$.

$$\mathbb{E}[R|A] = \sum_{x \in A} x \Pr[R = x|A] = \frac{1}{4}[1 + 2 + 3 + 4] = \frac{5}{2}$$

2. Law of Total Expectation

If R is a random variable $S \rightarrow V$ and events $A_1, A_2, \dots, A_n$ form a partition of the sample space (i.e. for all i, j , $A_i \cap A_j = \emptyset$ and $A_1 \cup A_2 \cup \dots \cup A_n = S$), then,

$$\mathbb{E}[R] = \sum_i \mathbb{E}[R|A_i] \Pr[A_i].$$

3. Independence of random variables

- **Definition**: Random variables R_1 and R_2 are independent if and only if for all $x_1 \in \text{Range}(R_1)$ and $x_2 \in \text{Range}(R_2)$, events $[R_1 = x_1]$ and $[R_2 = x_2]$ are independent. Formally,

$$\Pr[(R_1 = x_1) \cap (R_2 = x_2)] = \Pr[R_1 = x_1] \Pr[R_2 = x_2]$$

Alternate definition of independence

Random variables R_1 and R_2 are independent if and only if for all $x_1 \in \text{Range}(R_1)$ and $x_2 \in \text{Range}(R_2)$,

$$\Pr[(R_1 = x_1) \mid (R_2 = x_2)] = \Pr[R_1 = x_1]$$

$$\Pr[(R_2 = x_2) \mid (R_1 = x_1)] = \Pr[R_2 = x_2]$$

Pairwise independence

Similar to events, random variables $R_1, R_2, \dots, R_n$ are pairwise independent if and only if for any pair R_i, R_j for all $x \in \text{Range}(R_i)$ and $y \in \text{Range}(R_j)$, events $[R_i = x]$ and $[R_j = y]$ are pairwise independent implying that

$$\Pr[(R_i = x) \cap (R_j = y)] = \Pr[R_i = x] \Pr[R_j = y]$$

Alternatively, R_i and R_j are pairwise independent if and only if for all $x \in \text{Range}(R_i)$ and $y \in \text{Range}(R_j)$,

$$\Pr[(R_i = x) | R_j = y] = \Pr[(R_i = x)]$$

$$\Pr[(R_j = y) | R_i = x] = \Pr[(R_j = y)]$$